

# Waffy Ahmed

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## Education

**Georgia Institute of Technology** | Atlanta, GA

Bachelor of Science in Computer Science

December 2024

Faculty Honors, Dean's List, Zell Miller Scholar

## Experience

### Software Engineer

*The Home Depot*

January 2025 – Present

*Atlanta, GA*

- Increased system resilience by implementing Kubernetes Horizontal Pod Autoscaling (HPA) for a core service previously capped at 20 static pods, reducing mean latency from 121 to 72 ms and error rates by 89% while increasing throughput from 24.9M to 34.8M requests/week and enabling dynamic scaling up to 100 pods.
- Reconstructed deployment workflows for legacy Java 1.8 Tomcat services lacking operational ownership by rebuilding CI/CD pipelines and enabling zero-downtime Cassandra credential rotation, eliminating a platform outage risk capable of halting all transaction capture.
- Led testing and validation to disable a legacy service component handling ~27% of transaction volume (14.5 million daily captures/lookups) using BigQuery and Postman, supporting a phased retirement effort involving 12 teams.
- Reduced mean time to recovery ~50% by implementing an automated Kubernetes rolling-restart job for 20+ microservices, replacing slow Argo Rollouts and accelerating recovery from memory-leak/state-drift issues.
- Automated Spinnaker deployment pipelines for 15+ microservices with cdk8s, standardizing multi-environment release workflows using TypeScript, Terraform, and GCP, reducing deployment times by 25%.

### Software Engineer Intern

*The Home Depot*

May 2023 – July 2023, May 2024 – July 2024

*Atlanta, GA*

- Developed an internal product information page for cashiers, streamlining self-checkout processes and reducing customer wait times for associate interventions by ~25%.
- Leveraged Java, React, and TypeScript to retrieve and surface validated product data across all self-checkout registers, mitigating customer theft by ~\$750,000 annually.
- Optimized the efficiency of configuring 40,000+ registers across 2,300 stores by 90%, in collaboration with 2 interns.
- Truncated maximum deployment time for register updates from 24+ hours to 30 minutes by using Java HttpClient and Flow APIs to streamline retrieval of polymorphic external configurations.

### Firmware Engineer Intern

*Landis+Gyr*

May 2022 – August 2022

*Alpharetta, GA*

- Accelerated conversion of hexadecimal data control values to ASCII representations by ~50% via Python logic.
- Bolstered data transmission efficiency by ~25% by collaborating with a software architect to debug Python-based collector systems and validate MySQL-backed data pipelines, improving reliability and processing performance.

## Projects

### Fintech @ Georgia Tech | *React, React Native, TypeScript, Stripe API, Chakra UI*

- Reduced page load times by ~35% via transforming a multi-page credit card website into a React-based single-page application, streamlining card wallet management and adding a card removal feature for authenticated users.
- Implemented a cart page for a React Native grocery application, allowing users to add/remove items, adjust quantities, view real-time total costs, and integrated the Stripe API for streamlined in-app credit card payments.

### CDC Data Reconciliation | *Python, FastAPI, SQLite, React, Tailwind CSS*

- Led a team of 6 to automate the reconciliation of case counts for infectious diseases between 50 state health departments and the CDC, alongside doubling as a full-stack software developer.
- Utilized Python to compare state health department and CDC data, highlighting any discrepancies between the two datasets, saving 5000+ hours of manual reconciliation annually.

## Skills

**Programming Languages:** Java, Python, JavaScript, TypeScript, SQL

**Technologies/Frameworks:** Kubernetes, GCP, Terraform, Cassandra, Elasticsearch, BigQuery, Google Firebase, Git, React, React Native, Node.js, Next.js, Express.js, SQLite, Postman, Docker, JUnit, Jest, Prometheus, PromQL

**Concepts:** Object-Oriented Programming & Design, Test-Driven Development, NoSQL, Data Structures & Algorithms, Full-Stack, CI/CD, Infrastructure, Distributed Systems, Cloud Computing, Microservices Architecture